

DUC-DUY VU

Full-stack Protein Designer

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WORKING EXPERIENCE

1/2025 – present: Full-stack Protein Designer

Computational Structural Biology Unit, Institut Pasteur, Paris, France

- Implementing state-of-the-art protein design pipelines to create protein binders for important therapeutic targets.
- Developing a new pipeline for designing protein interaction hubs that can bind simultaneously to two partners.
- Developing a new pipeline for designing proteins that can change conformation upon phosphorylation.
- Testing > 100 designs in the lab using various biophysical techniques (CD, BLI, NMR).
- Conducting large-scale structural protein predictions and creating protein viewer dashboards to generate new biological hypotheses.
- Developing a geometric deep learning model to classify and cluster short linear interacting motifs (SLiMs)

09/2023 – 12/2024: Integrative Structural Biology Study of Disordered Proteins

Laboratoire des biomolécules, UMR 7203, Paris, France

- Revealed the molecular interactions underlying the liquid-liquid phase separation of DNA double-strand break repair by NHEJ machinery.
- Elucidated the binding mechanism of an intrinsically disordered protein using chemical exchange NMR.
- Characterized the nanosecond dynamics of intrinsically disordered proteins using high-resolution NMR relaxometry.
- Established and optimized protocols for labeling fluorine in proteins for NMR studies.
- Developed new shuttle-NMR experiments to characterize high molecular weight proteins.
- Supervised two PhD students and three master students.

01/2019 – 06/2019: Protein Biochemist

Institute of Structural Biology, Grenoble, France

Characterization of the interaction between nucleoprotein and phosphoprotein in Rabies virus replication machinery using biophysical techniques combined with molecular dynamics simulation

12/2017 - 07/2018: Computed-aided Drug Design

University of Medicine and Pharmacy at Ho Chi Minh City, Viet Nam

Design, synthesis, *in silico* characterization (molecular docking and molecular dynamics simulation) and *in vitro* biological assay of pyrrolidine/piperidine chalcone derivatives as acetylcholinesterase inhibitors.

EDUCATION

PhD in Integrative Structural Biology

Ecole Normale Supérieure – PSL university, Paris, France

- Thesis project: “[Functions and Dynamics of Intrinsically Disordered Protein Regions in Non-homologous End-Joining by NMR spectroscopy](#)”, supervisor: Dr. Fabien FERRAGE
- Supervising 3 master students and 2 bachelor students

Master in Integrative Structural Biology

Université Grenoble Alpes, Grenoble, France

- Fully-funded IDEX scholarship
- GPA: 16.963/20 - Very good (Rank: 1st)

Bachelor in Pharmaceutical Science

UMP, Ho Chi Minh city, Viet Nam

- Tuition scholarship for top students, GPA: 7.41/10 - Good
- Training in a wide range of knowledge, from fundamental sciences to clinical subjects

RESEARCH SKILLS

- **Computational structural biology:** Protein structure modeling, high-throughput protein-protein interaction prediction, binder design, and sequence redesign to improve stability/solubility by deep learning (AlphaFold, Boltz, RFDiffusion, BindCarft, proteinMPNN, Pyrosetta).
- **Biophysics:** Isotope labelling of proteins (uniform or selective) for structural characterization, NMR assignment, and characterizing the kinetics and dynamics of protein/DNA-protein interactions using various approaches (NMR titration, 2D lineshape analysis, PRE, CPMG, CEST), NMR relaxometry, ITC, FP, MTS, DSC.
- **Biochemistry:** Extensive experience in protein expression and purification in bacteria, including the optimization and trouble-shooting of protocols; proficient in using the AKTA system for affinity, ion-exchange, and gel filtration chromatography; protein analysis using SDS-PAGE, Western blot, and SEC-MALLS; experienced in fluorescent protein labeling and utilizing epifluorescence microscopy to monitor liquid-liquid phase separation (LLPS), protein quality control (MS, DSL, UV-VIS).
- **Molecular biology:** PCR, site-directed mutagenesis, cloning techniques

COMPETENCE

Languages

- Vietnamese: mother tongue
- English: full professional proficiency
- French: sufficient working proficiency

Technical skills

- Python, Linux-based environments
- Machine learning/Deep learning with Pytorch
- Data analysis with Numpy and Pandas
- Data visualization with Matplotlib, Seaborn, Plotly, Dash

SCIENTIFIC PUBLICATION

Publications:

- "Multivalent interactions of the disordered regions of XLF and XRCC4 foster robust cellular NHEJ and drive the formation of ligation-boosting condensates in vitro" **Duc-Duy Vu et al.**, **Nature Structural & Molecular Biology** **31.11 (2024): 1732-1744.**
- "Structure and dynamics of the unassembled nucleoprotein of rabies virus in complex with its phosphoprotein chaperone module" Francine ..., **Duc Duy Vu**, et al. **Viruses** 14, no. 12 (2022): 2813.

Reviewer for Nature Communication, Nature Biotechnology

REFERENCES

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